

Experiment Assisting System with Local Augmented Body (EASY-LAB) for Subject Experiments under the COVID-19 Pandemic

Yukiko Iwasaki*
Waseda University
Shinjuku, Tokyo, JAPAN
kamiwaza@ruri.waseda.jp

Joi Oh*
Waseda University
Shinjuku, Tokyo, JAPAN
joy-oh0924@akane.waseda.jp

Takumi Handa*
Waseda University
Shinjuku, Tokyo, JAPAN
taku.handa@ruri.waseda.jp

Ahmed AlSereidi*
Waseda University
Shinjuku, Tokyo, JAPAN
ahmed@akane.waseda.jp

Vitvasin Vimolmongkolporn*
Waseda University
Shinjuku, Tokyo, JAPAN
vitvasin.au@fuji.waseda.jp

Fumihiko Kato*
Waseda University
Shinjuku, Tokyo, JAPAN
fumihiko.kato@aoni.waseda.jp

Hiroyasu Iwata*
Waseda University
Shinjuku, Tokyo, JAPAN
jubi@waseda.jp

ABSTRACT

Since it is challenging to perceive space and objects with a video conferencing system, which communicates using only video and audio, there are difficulties in testing subjects in the COVID-19 pandemic. We propose the EASY-LAB system that allows an experimenter to perform observation and physical interaction with the subject even from a remote location actively. The proposed system displays the camera image on a HMD worn by the experimenter, which camera is mounted on a small 6 DOF robot arm end, allowing observation from an easy-to-see perspective. The experimenter can also instruct the subject using another robot arm with a laser pointer. The robot's joint angles are calculated by Inverse Kinematics from the experimenter's head movements, then reflected in the actual robot. Photon Unity Networking component was used for the synchronization process with remote locations. These devices are affordable, effortless to set up, and can be delivered to the subject's home. Finally, the proposed system was evaluated by four subjects. As a preliminary result, the mean pointing error was 1.1 cm, and the operation time was reduced by 60% compared with the conventional video conferencing system. This result indicated the EASY-LAB's capability, at least in tasks that require pointing and observation from various angles. The statistical study with more subjects will be conducted in the follow-up study.

*The authors contributed equally to this research.

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CCS CONCEPTS

• Human-centered computing → Virtual reality; • Applied computing → Telecommunications.

KEYWORDS

teleexistence, telepresence, telecommuting work, UI/Tools, Interaction

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1 INTRODUCTION

Subject experiments are difficult to be conducted when a experimenter and a subject can not be in the same place. Researchers and engineers often ask subject to perform some task, observe its behavior, or use new devices and evaluate those usability. In these situations, advanced instructions to the subject given by words or videos are sometimes not enough to precisely inform the experimenter's intention. When the subject have a question or makes an error, physical interaction by the experimenter, such as teaching by actually holding hands or pointing some objects, can be useful. In addition, for this purpose, experimenter has to observe the subject's behavior from any point of view to observe the process throughout the experiment. However, these active observations and physical interventions are difficult if the experimenter and the subject are not in the same location at such the time of the COVID-19 disaster. Therefore, we aim to develop a new system that enables the experimenter to observe and physically intervene with the subject as if the experimenter was standing beside the subject, even if the subject are in a remote location.

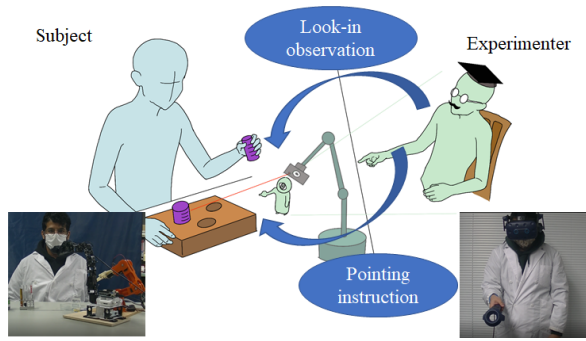


Figure 1: EASY-LAB concept

There are two essential requirements in the remote subject experiment system. One is that the experimenter should be able to intervene in the physical space of the subject actively. In recent years, tools such as video conferencing systems [Zoo 2021] [Goo 2021] are widely spread due to COVID-19 pandemic. They enable real-time exchange of information even in remote areas with clear images and sounds. On the other hand, this way of passing information does not allow for direct interaction with the physical space of the subject. In other words, there is some information that is difficult to convey with such methods. For example, instructing "a little more to the right" or "assemble that red object into the next hole" is easier to understand intuitively by pointing directly at its position rather than using many instruction words. Besides, in most videoconferencing systems, the receiver side fixes the camera. Therefore, the experimenter cannot actively move his or her head to observe the subject.

Several systems [Kuzuoka et al. 1994] [Kurata et al. 2004] have been developed in the past with the objective of moving a robotics device using a natural gesture humans constantly use. Researches on tele-operated robots has been going on for a long time with many proposals for full-body immersion systems [Tachi et al. 2020] and elaborate systems that can reproduce the movement of the fingers of a hand [Yamano and Maeno 2005][Butterfass et al. 2001]. However, these systems are not suitable for speedy and general implementation as they had some issues such as being difficult to distribute, high cost, and usability from a remote location is unstable. The system ought to be usable with minimum latency from a distant location, easy to set up in a small space, and with low cost. For example, depending on the nature of the experiment, it is sufficient to hold a lightweight object such as a wooden block or point at a certain point with a laser. In order to resume subject experiments that had to be stopped or changed to alternative methods because of the pandemic, having a simple and cost-efficient system is the best way to solve this issue, since VR headsets are widely available, setup for the instructor is easier. Previous works usually required a specific combination of sensors.

To solve this, we propose the Experiment Assisting System with Local Augmented Body (EASY-LAB, Figure 1) to support subject experiments, which can be installed easily and allow experimenters to observe and physically interact with the subjects actively. We divided the subjective experiments into different categories depending on the required level of physical intervention needed. In this paper, we primarily focus on developing the prototype up to Level 1

and studying the usability of the system. The Level of the prototype is described as follow:

- Level 0** - Active observation; for cognitive task or psychophysical experiments which require observation of subjects.
- Level 1** - Active observation and intervention with pointing; for manual work with assembling or picking and placing something that requires direct pointing for precise instructions.
- Level 2** - Active observation and intervention with simple haptic robot hand; for wearing devices or instruments which require fine-tuning of position or to check its tightening.

2 METHOD

2.1 System Requirements

There are two functional requirements to achieve Level 1 of EASY-LAB, observation, and pointing. For observation, it is necessary that the experimenter can look into the subject from any angle. This function can be done, for example, by using a controller to set the position and angle of the camera installed on the subject's side. A more intuitive method would be to reflect the movement of the experimenter's head directly to the camera's movement. In this case, we need a system that can measure the angle and position of the experimenter's head and a robotic arm with at least 6 DOFs to mount the camera.

For pointing, it must visually indicate to the subject any point that the experimenter points at. One of the most straightforward solutions is to set up a laser pointer on the subject's side. The movement of the pointer can be synchronized to the experimenter's head or hand movement. If the laser is synchronized with the head movement, the system can be made compact because only one robot arm is used to control both the camera and the laser. According to previous studies [Iwasaki and Iwata 2018][Oh et al. 2020], experimenters can point simply by moving their head only. In selecting the hand movement, its main advantage is that observation and instruction can be done independently. To achieve this, another robot arm to mount a laser pointer is required. A robot arm requires at least 3 DOFs to point, but if we try to keep the system small or generate a continuous trajectory to give instructions (e.g., drawing a circle to show the area), a robot arm with up to 6 DOFs is required.

2.2 System Configurations

Figure 2 shows the system configuration of EASY-LAB. It consists of three sections. The experimenter section acquires the head or hands' position and angle using VIVE. The subject section is equipped with robot arms with a camera and a laser pointer attached to them. Lastly, The network section links these two sides to transmit information using the Photon Unity Networking system.

2.2.1 Experimenter Section. This section consists of the immersive head mounted display and computer. A VIVE system is used for this, position and rotation of the experimenter's head or hand are sent to the computer using Unity. The experimenter can see the camera view from the subject's section in the head mounted display. Using robot arm simulation in Unity, solve the joint angle with IK to matches the position and posture (6 DOFs) of the experimenter's head, and synchronize the tip of the actual robot.

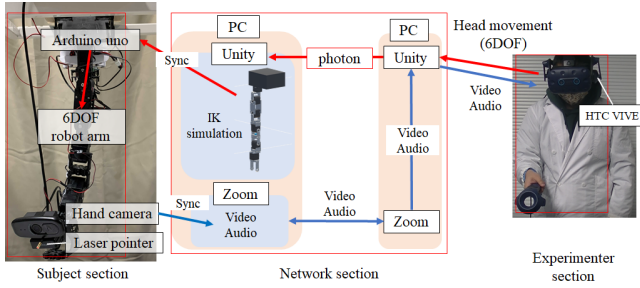


Figure 2: System configuration of EASY-LAB level 1 prototype

2.2.2 Subject Section. This section consists of a robot arm mounted with the camera and a laser pointer. The robot arm that is mounted with a camera, we used a KXR-A5 Ver.2 (6 DOFs, KONDO KAGAKU Co.,Ltd.) for this is to reproduce facial movement and at the same time enable peeking movement. Considering its torque, we used the camera BSW500MBK (110 g, BUFFALO Inc.). We prepared another robot arm Tinkerkit Braccio (4 DOFs, Arduino.org produced) for mounting the laser pointer, and both robot arms are controlled by Arduino [Ard 2021]. For the laser pointer, we used RED dot laser module RP650AD5-4C (Class 2, Changchun Realpoo Photoelectric Co., Ltd.). The Class 2 laser satisfies the JIS C 6802 and can be switched off, except when the experimenter is using it. However, as a further safety measure, we installed a safety switch on the subject's side (robot arm side). Therefore, the robot arm can be switched off immediately if the subject feels uncomfortable. In the Level 1 experiment, a laser pointer was attached to the robot arm of the camera. The total cost of the system (when excluding the pointer-mounted robot arm) was about 641 USD.

2.2.3 Network Section. This section consists of the application made by Unity and Photon Unity Networking [PUN 2021]. Here, the positions and angles of the head or hands sent from the experimenter section are received as the positions and angles of virtual objects in Unity, and the subject section reads those information as input values to control the robot. These platform engines are provided free of charge and help reduce the overall system cost, but they are not suitable for footage sharing quickly. Therefore, to effectively exchange the video footage, we used a typical video conferencing application such as Zoom [Zoo 2021] or Google Meet [Goo 2021]. The network section captures the screen that has been provided by these external applications and displays them on the Unity screen, thereby reducing communication costs and allowing the experimenter and subject to use the EASY-LAB together with tools they are familiar with.

3 USER STUDY

3.1 Overview

In this chapter, we verify the usability of the developed system through a user study. The requirements of the system were quick and easy setup and the ability for the experimenter to observe and give more direct instructions. The system's setup is a laser pointer mounted on the camera-mounted robot arm from the discussion in the previous chapter. As a pilot study, we asked the cooperation

of four robotics researchers (mean age 24.75, SD 1.30, 3 male and 1 female) who were previously involved in subject experiments. First, we asked the participants to set up the robot arm from the state that the arm was in the box (Figure 3(a)) and the PC was activated, based on the instruction paper. Second, we evaluated the accuracy of pointing and the satisfaction of communication through a simple task.

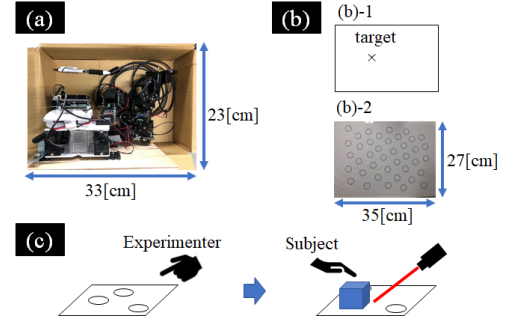


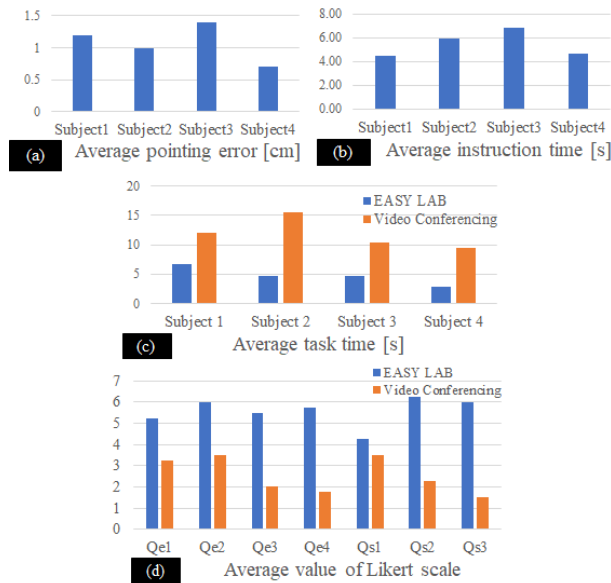
Figure 3: Task and procedure: (a)EASY-LAB packaging kit, (b)task sheet, (c)task procedure

3.2 Task and Procedure

First, a sheet in the Figure 3 (b)-1 was prepared on the subject's side, and the experimenter was asked to point at the target point from 13 cm away of the initial point. When the experimenter says "done", the error distance of pointing instruction was measured, which was repeated 3 times. Next, separated into two conditions, using EASY-LAB and a conventional video conferencing system [Zoo 2021], subjects performed a task in which they placed a block in a given location as instructed by the experimenter (Figure 3(c)). This task simulates the assembly work, which requires the experimenter to direct the object in that position and then confirms that it is completed by observing it from multiple angles. As shown in the Figure 3 (b)-2, the target circle is randomly placed in an area of 27 x 35 cm, and the subject must quickly place a wood block so that the designated circle is hidden. The correct arrangement of the block is known only to the experimenter just before the trial. The task is completed when the experimenter deems that the block is placed in their designated circle. After the completion of the task, the subjective evaluation was conducted using a questionnaire (contents are shown in the Table 1) using a 7-point Likert scale and interviews. All four participants experienced the subject's side and the experimenter's side once in random order.

3.3 Result and Discussion

The mean pointing error between the participants was 1.1 cm, and the average time to pointing was 5.48 s. The individuals average pointing errors are shown in Figure 4-(a) and the times are shown in Figure 4-(b). Figure 4-(c) and Figure 4-(d) show the time taken to complete the task and the scores of each questionnaire. Since we have only a few participants at this point, we will leave the statistical analysis for a future follow-up report. However, all participants improved their task performance when using the EASY-LAB and recorded higher scores in the subjective evaluation. This result suggests EASY-LAB's effectiveness, at least in tasks that require pointing and observation from multiple angles. On the other hand,

Table 1: Subjective evaluation questionnaire**Experimenter's side****Qe1** I felt like I was in the same space as the subject.**Qe2** I could see the subject's work well.**Qe3** I was able to convey my intention to the subject with confidence.**Qe4** I could communicate my intention to the subject quickly.**Subject's side****Qs1** I felt as if the experimenter was in the same space.**Qs2** I was able to work confidently with the experimenter's instructions.**Qs3** I was able to understand the experimenter's intention immediately.**Figure 4: User study results**

the Photon Network experienced a delay of 0.05 s, and even more delay was assumed due to the encoding in the web camera. Regarding delays, some participants mentioned it as a concern in the interview. Improving responsiveness is a future technical challenge. With the current prototype, it is tough to perform tasks that require precise timing instructions (e.g., pouring water and stopping when a certain amount is reached). As for the presence of subjects and experimenters in each other (Qe1 and Qs1), the results showed significance between the conventional system and EASY-LAB, but there is room for discussion on whether it is better to increase or decrease this presence. For example, in psychophysical experiments, there are concerns about the effects of the experimenter's guidance and atmosphere on the result of the subject's experiment [Cobos et al. 2015][Kintz et al. 1965]. If we can create a system that allows the experimenter to feel the subject's presence and the subject to be less aware of the presence of the experimenter, it could be a new solution to these problems.

4 CONCLUSION

In this study, we proposed EASY-LAB's concept to enable subject experiments in situations where the experimenter and the subject

cannot be in the same place as in the COVID-19 pandemic. We developed a prototype system that enables looking-in observation and pointing instruction using a robot arm, an immersive head mounted display. After that, we conducted a user study using the simple task. As a result, the system achieved satisfying results of being affordable (641 USD) and easy setup (101 s). It received higher subjective ratings than conventional video chat instruction. The main contribution of this work is to propose the simple and low-cost prototype for resuming subject experiments from remote place under the COVID-19 pandemic. For future works, we will soon advance to developing the prototype of level 2. Next challenge is to develop the haptic controller as simple configuration as possible. For example, we try to produce the sense of tightening or texture by small tactile module [Nakatani et al. 2018]. Also, we intend to prepare further dissemination of the system by developing a small 6 DOF robot arm that can mount a personal smartphone or other devices to replace the current camera and release an application that can be installed to it.

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REFERENCES

- Online; accessed May-2021. Arduino. <https://www.arduino.cc/>.
- Online; accessed May-2021. Google Meet. <https://apps.google.com/intl/ja/meet/>.
- Online; accessed May-2021. Photon Unity Networking. <https://www.photonengine.com/ja/PUN>.
- Online; accessed May-2021. Zoom. <https://zoom.us/jp-jp/meetings.html>
- J. Butterfass, Markus Grebenstein, Hangzi Liu, and G. Hirzinger. 2001. DLR-Hand II: next generation of a dextrous robot hand. In *Proceedings 2001 ICRA. IEEE International Conference on Robotics and Automation (Cat. No.01CH37164)*, Vol. 1. 109–114 vol.1. <https://doi.org/10.1109/ROBOT.2001.932538>
- Briana Cobos, Kelly Zolnierok, and Krista Howard. 2015. White coat hypertension: Improving the patient–health care practitioner relationship. *Psychology research and behavior management* 8 (05 2015), 133–41. <https://doi.org/10.2147/PRBM.S61192>
- Yukiko Iwasaki and Hiroyasu Iwata. 2018. A face vector - the point instruction-type interface for manipulation of an extended body in dual-task situations. In *2018 IEEE International Conference on Cyborg and Bionic Systems (CBS)*. 662–666. <https://doi.org/10.1109/CBS.2018.8612275>
- D.J. Kintz, B.L. and Delprato, Persons C.E. Mettee, D.R., and R.H. Schappe. 1965. The experimenter effect. *Psychological Bulletin* 63, 4 (1965), 223–232. <https://doi.org/10.1037/h0021718>
- T Kurata, N Sakata, M Kourogi, H Kuzuoka, and M Billingham. 2004. Remote collaboration using a shoulder-worn active camera/laser. In *Eighth International Symposium on Wearable Computers*, Vol. 1. IEEE, Los Alamitos CA, 62–69.
- Hideaki Kuzuoka, Toshio Kosuge, and Masatomo Tanaka. 1994. GestureCam: a video communication system for sympathetic remote collaboration. In *Proceedings of the 1994 ACM conference on computer supported cooperative work (CSCW '94)*. ACM, 35–43.
- Masashi Nakatani, Katsunari Sato, Kunio Sato, Yuzuru Kawana, Daisuke Takai, Kouta Minamizawa, and Susumu Tachi. 2018. A Novel Multimodal Tactile Module that Can Provide Vibro-Thermal Feedback. In *Haptic Interaction*, Shoichi Hasegawa, Masashi Konyo, Ki-Uk Kyung, Takuya Nojima, and Hiroyuki Kajimoto (Eds.). Springer Singapore, Singapore, 437–443.
- Joi Oh, Koza Ando, Shuhei Iizuka, Lena Guinot, Fumihiro Kato, and Hiroyasu Iwata. 2020. 3D Head Pointer: A manipulation method that enables the spatial localization for a wearable robot arm by head bobbing. In *2020 23rd International Symposium on Measurement and Control in Robotics (ISMCR)*. 1–6. <https://doi.org/10.1109/ISMCR51255.2020.9263775>
- Susumu Tachi, Yasuyuki Inoue, and Fumihiro Kato. 2020. TELESAR VI: Telexistence Surrogate Anthropomorphic Robot VI. *International Journal of Humanoid Robotics* 17, 05 (2020), 2050019. <https://doi.org/10.1142/S021984362050019X>
- Ikuo Yamano and Takashi Maeno. 2005. Five-fingered Robot Hand using Ultrasonic Motors and Elastic Elements. In *Proceedings of the 2005 IEEE International Conference on Robotics and Automation*. 2673–2678. <https://doi.org/10.1109/ROBOT.2005.1570517>